
Dahan River Waterfront Facilities Management Dashboard System Plan

System Specification

Requirement Definition

This system is the Dahan River Waterfront Facilities Management Dashboard. Built on open-source and government open data, it integrates GeoJSON layers for pumping stations, water level stations, and rivers. By connecting real-time CWA weather/rainfall data, New Taipei City township forecasts (F-D0047-071), and real-time water levels from the Water Resources Agency, the system allows users to click on the map to view nearby station weather and basin risk assessments. The system uses WGS84 coordinates, with GeoJSON and REST JSON as the primary formats for data exchange.

Development Environment

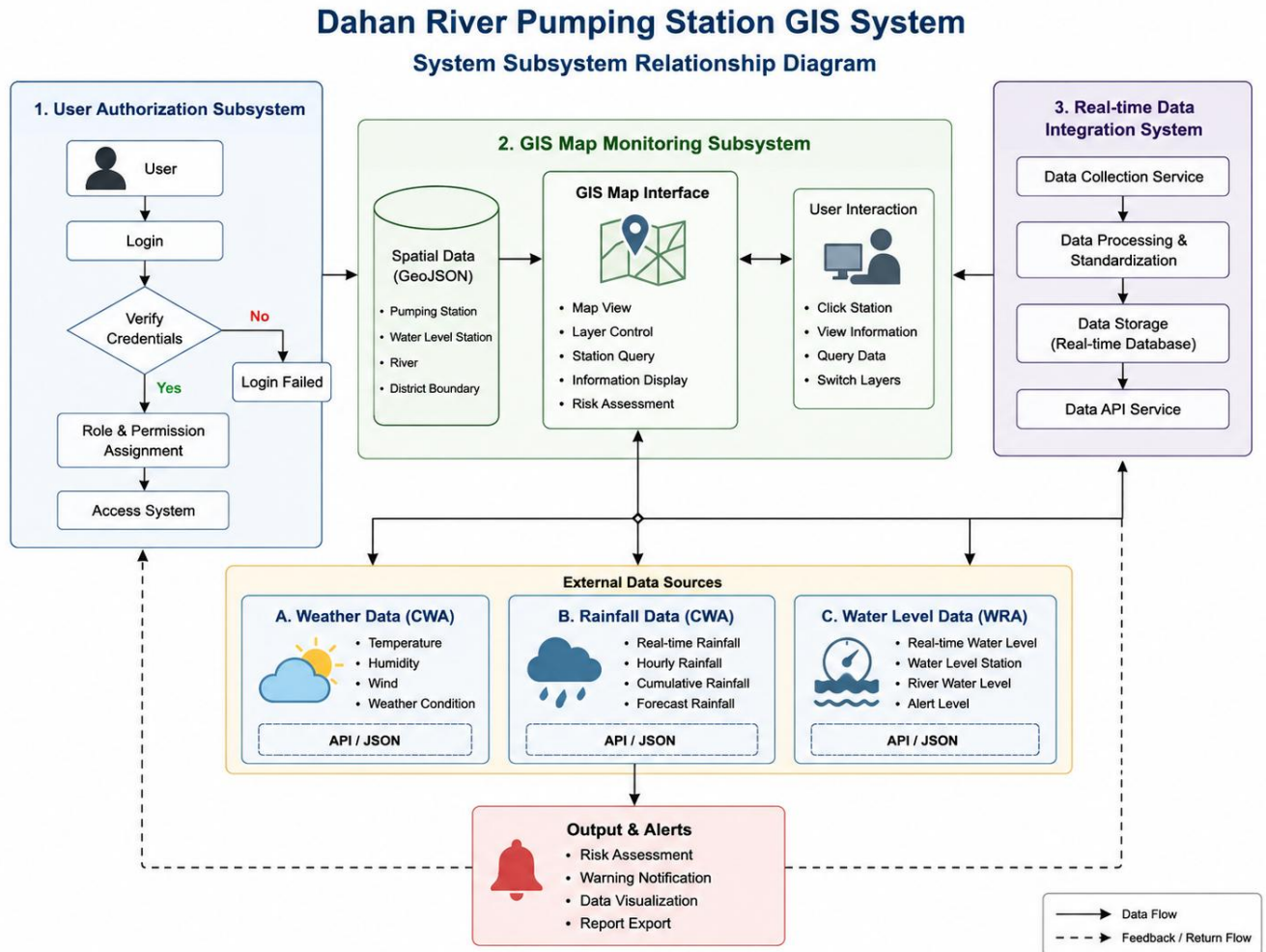
- Target web browsers:
 - Google Chrome
 - Microsoft Edge
 - Apple Safari
- Development tools
 - Coding: HTML5、CSS3、JavaScript、Node.js、Express
 - Web-based GIS map view
 - Google Maps JavaScript API（互動式底圖、標記、資訊視窗與點選互動）
 - GeoJSON 向量圖層（抽水站、水位站、河川、河岸廊道）
- Data Standard
 - Spatial Layer Format: GeoJSON (WGS84)
 - Coverage: Dahan River basin and relevant administrative districts in Taipei and New Taipei City
 - Data Integration: Adheres to government open data policies and OGC-compliant vector data and API integration methods

System Scope

Dahan River Waterfront Facilities Management Dashboard System includes 3 subsystems as below:

1. GIS Map Monitoring System
2. User Authorization
3. Real-time Data Integration System System...

The main functions and data flow for each subsystem are planned according to system requirements.



System Development Plan

The system development plan focuses on developing a GIS-based monitoring and management platform for pumping stations in the Dahan River Basin. The system integrates spatial information, real-time weather data, rainfall data, and water level monitoring services to support facility management, risk assessment, and decision-making processes.

The development plan includes system structure design, user interface design, data integration mechanisms, and user authorization management. The system provides an interactive GIS environment that enables users

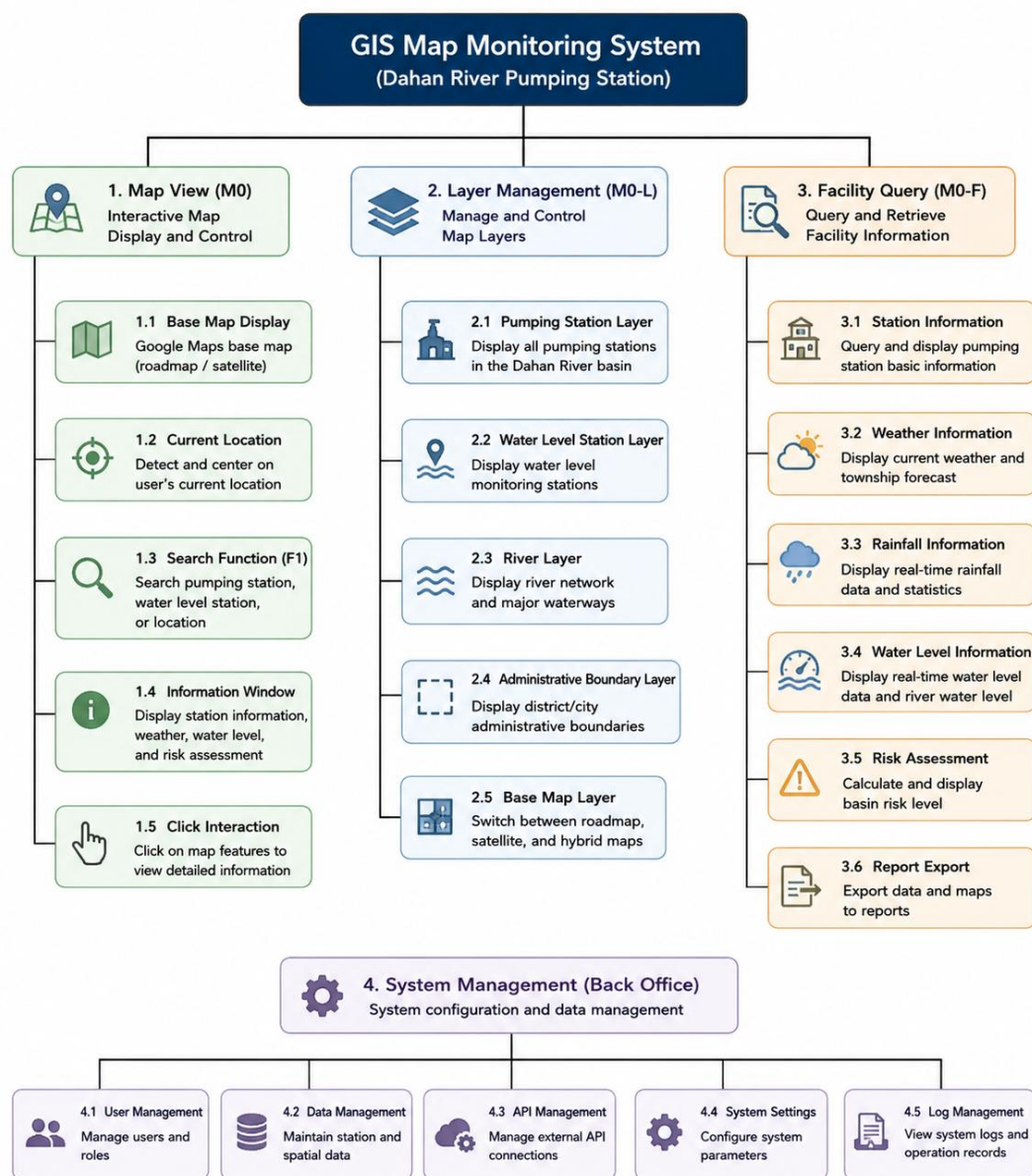
to visualize pumping stations, monitor environmental conditions, and access real-time operational information.

The following subsystems are included in the development plan:

1. GIS Map Monitoring System
2. User Authorization System
3. Real-time Data Integration System

GIS Map Monitoring System

System Structure



Conceptual UI Elements



Main Functions

The main map view is shown when the user is logged in. Users can browse pumping stations, water level stations, rivers, and related geographic information through the GIS map interface.

1. Map View (M0)

The main map view is shown when the user is logged in. By default, view M0-T is shown; for phones, it will cover this view, but for tablets, it will take up a small portion of the right part of the screen.

- UI Elements

M0-01	Search button – Search pumping stations, water level stations, or locations.
M0-02	Current location button – Centers the map on the user's current location.
M0-03	Layer selection button – Opens the layer management menu.
M0-04	Facility information window – Displays detailed station information when a map marker is selected.

2. Layer Management (M0-L)

Popup menu for selecting map layers displayed in the GIS map. Users can enable or disable layers according to operational requirements.

- Available independent layers:

Pumping Station Layer	Displays all pumping stations.
Water Level Station Layer	Displays water level monitoring stations.
River Layer	Displays river and waterway information.
Administrative Boundary Layer	Displays administrative boundaries.

- Available Base Maps:

Road Map	Standard Google Map view.
Satellite Map	Satellite imagery view.
Hybrid Map	Combination of road map and satellite imagery.

3. Facility Information Query (M0-F)

This view appears when users select a station marker on the map. The system retrieves station-related information and displays it in the information window.

- UI Elements

M0-F01	Pumping Station Information – Displays station name, location, and basic information.
M0-F02	Weather Information – Displays current weather conditions and forecasts.
M0-F03	Water Level Information – Displays real-time water level data.

M0-F04

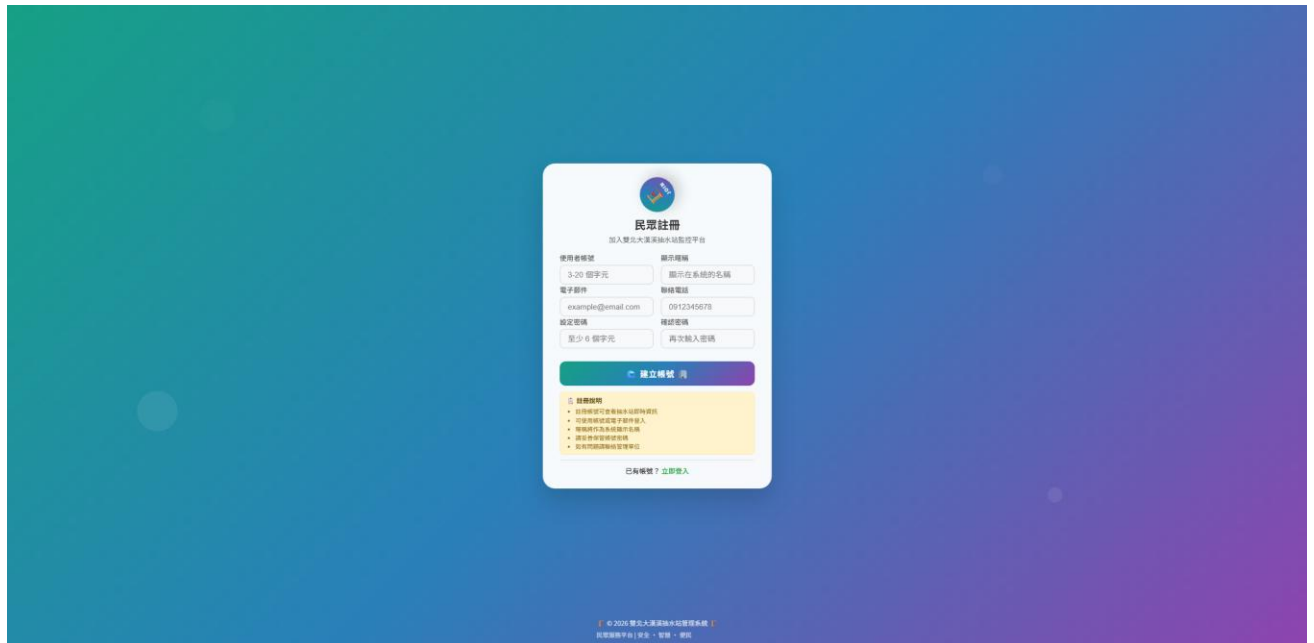
Rainfall Information – Displays rainfall data and statistics.

M0-F05

Risk Assessment Information – Displays risk assessment results for the selected area.

User Authorization

Conceptual UI Elements



使用者註冊介面



使用者登入介面

Main Functions

The login view is the first view users see when accessing the system. During login, the server verifies user credentials and determines user roles and permissions.

1. Login (L0)

● UI Elements

L0-01	System Title
L0-02	Username Input Box
L0-03	Password Input Box (password will be masked)
L0-04	Error Message Area
L0-05	Login Button

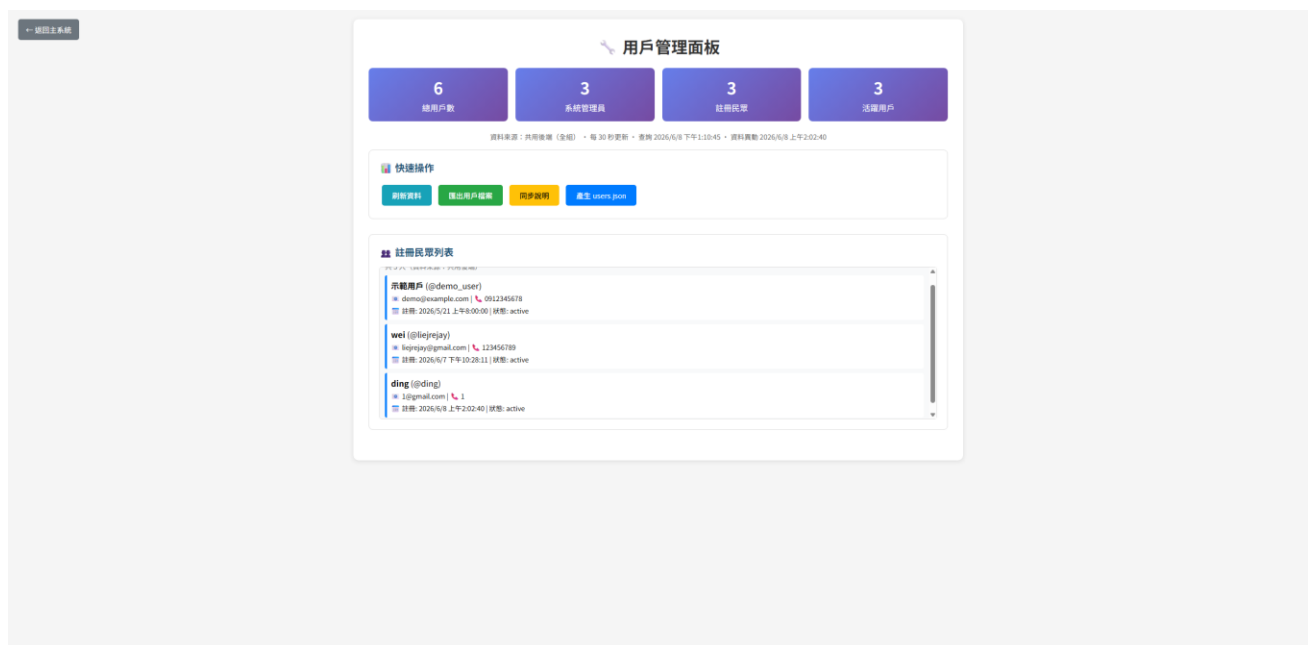
Possible error messages for L0-02:

Message ID	Message Text
L0-M01	請填寫這個欄位。
L0-M02	帳號或密碼錯誤。

2. Role Management (L1)

The system assigns permissions according to user roles after successful login.

Role	Description
Administrator	Full access to all system functions and user management.
Viewer	Read-only access to monitoring information.



管理者管理面板(僅管理者可見)